

Natalia Agapova

ML Engineer

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Applied ML: NLP, explainability and debiasing, computer vision, RAG. I ship models with metrics, ablations, and honest trade-off analysis — accurate, explainable, and product-ready. Published research; I connect models to interfaces.

EXPERIENCE

Croissan Studio · ML, design

Apr 2025 — present

AI products in a full-cycle studio: models, interfaces, and how results are presented.

Asimov Lab · ML, UX/UI, front-end

Feb — Sep 2024

AI test-generation startup; front-end team lead. Full path from data to UI.

Freelance · ML

2022 — present

Applied ML alongside design and web development.

SELECTED PROJECTS

Fair resume screening

NLP · [GitHub](#) · [paper](#)

BERT resume classifier for 9 IT supercategories (HeadHunter). Integrated Gradients (Captum) for bias audit; 39+ debiasing configs. Baseline: 60.9% accuracy, city-swap flipped predictions in 7.7% of pairs. Data scrubbing: 0% flip at 59.4% accuracy.

Emotion detection

CV · FER2013 · on-device

EmotionCNN: 58.7% accuracy, macro-F1 0.569; up to 60.8% with ensemble+TTA. INT8 quantization $\square$ Core ML and ONNX Runtime Web, 1.15 ms latency. Full loop: data, training, eval, deployment.

Gesture input

CV · MediaPipe · real-time

Webcam cursor control via pinch / ready / click gestures; PyAutoGUI, smoothing, 60 FPS. Rule-based classifier on 21 landmarks.

Personal Knowledge System

RAG · NLP · Python

Answers from personal PDFs grounded in context only, with filename+page citations. Chunking, SentenceTransformer, ChromaDB, Ollama llama3.2 — no LangChain, every layer hand-written.

PUBLICATIONS

Fair resume screening, BERT debiasing — DOI: 10.66693/mathai.1017.

Kholodov Y. A.; Salloum H.; Agapova N. A. Quantum annealing and RSA. *Voprosy kiberbezopasnosti*, 2025. DOI: 10.21681/2311-3456-2025-4-127-133.

SKILLS & EDUCATION

Python, PyTorch · BERT, Captum · debiasing, counterfactuals · RAG, ChromaDB, Ollama · MediaPipe, OpenCV · Core ML, ONNX · English C1+.

Innopolis University, BSc in Applied Artificial Intelligence, 2026.

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